Braxton Mira-Knippel

INLS 697

April 2026

Payroll Efficiency, Owner Wealth, and Championship Success in Major League Baseball: A Data-Driven Information Science Analysis of MLB Price per Win (PPW)

This paper analyzes Major League Baseball (MLB) payroll efficiency by using a relational dataset of all 30 MLB teams over a 7 year period (2019–2025). It examines which teams achieved the lowest average price per win (PPW) and whether having a low average PPW correlates with World Series championships. It also examines how owners' net worth and method of wealth acquisition (earned versus inherited) are related to payroll spending and team success. Drawing on INLS 523 (database normalization), STOR 320 (relationship modeling), and INLS 560 (information visualization), this study normalizes the dataset and applies data science techniques learned during four years in Information Science at UNC.

Major League Baseball teams individually spend hundreds of millions of dollars annually on player payroll, yet the efficiency of that spending, measured as the price per win (PPW), varies widely. Using the MLB payroll dataset found on Baseball Reference and cross referenced with Fangraphs, this paper discusses two questions: (1) Which teams pay the least per win on average, and does having a low PPW correlate with World Series championships? (2) How do MLB owners' net worth and method of wealth acquisition (earned vs. inherited) relate to payroll spending and team success?

These questions highlight an increasingly challenging problem in information science: applying relational databases and data modeling to professional sports and using that ability to improve teams. By treating the spreadsheet as a relational table and applying normalization principles (INLS 523), relationship modeling (STOR 320), and visualization techniques (INLS 560), this analysis shows how information science tools can show patterns invisible to most players, owners, and fans. My hypothesis is that spending efficiently (low PPW) is more valuable than spending as large an amount as possible. While recent World Series winners (2023–2025) spent above the true average PPW, the winners from 2019 to 2022 were all below average, suggesting that efficiency is a viable long-term driver of competitiveness and may be more valubale than high spending alone.

By treating the MLB payroll spreadsheet as a relational database, (Elmasri and Navathe (2017)) normalizing the raw data by separating entities such as Team, Avg PPW, Owner NET (net worth) , Owner Gain (how they gained wealth), and World Series into tables linked by primary

keys. By doing this, I was able to help preserve functional dependencies and avoid anomalies when analyzing the relationships. (INLS 523)

Modeling the relationships between variables (Stor 320) helped create a correlation analysis and a scatterplot that examined the connections between PPW, owner net worth, and championship outcomes. Using techniques for communicating the data clearly to others, I was able to ensure graphs highlight patterns without distortion or confusion. (INLS 560)

(dataset 1)

Elmasri and Navathe (2017) emphasized that proper normalization of relational databases is essential. I made sure this principle applied directly to the MLB dataset. Han, Kamber, and Pei (2012) describe data mining techniques that show hidden patterns in large relational tables, supporting the identification of PPW trends. Provost and Fawcett (2013) highlight the business value of data science models for predicting outcomes, leading to a correlation analysis between payroll efficiency and championships.

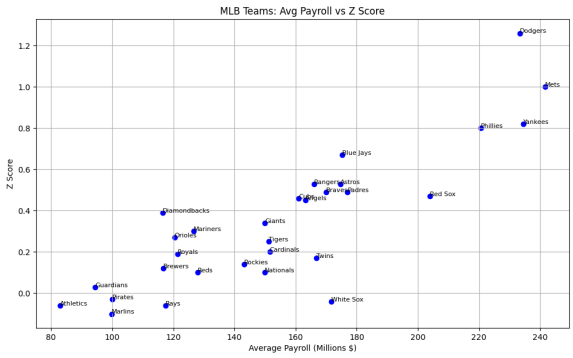
Floridi (2013) however, highlighted ethical issues around power imbalances when decisions are driven by data, something that applies to how team owners interpret payroll metrics. Similarly, Boyd and Crawford (2012) warned against buying into “big data” hype, instead stressing that context is required above all else. That’s especially relevant here when trying to figure out whether a low PPW actually predicts success, or if it’s just a reflection of larger market and spending limitations.

The dataset consists of 30 MLB teams with columns for Team, yearly payroll (2019–2025), Avg PPW, Owner name, Owner NET (net worth), Owner Gain (earned vs. inherited), Average Payroll, Z scores, and WS (World Series winner in the timeline). The spreadsheet was imported and normalized to create clean relational tables. Total wins combine regular-season and postseason results. True Avg (or Tr Avg) represents the mean of the mean and median spending for the year. Percent change shows how much more (or less) a team pays compared to the league average.

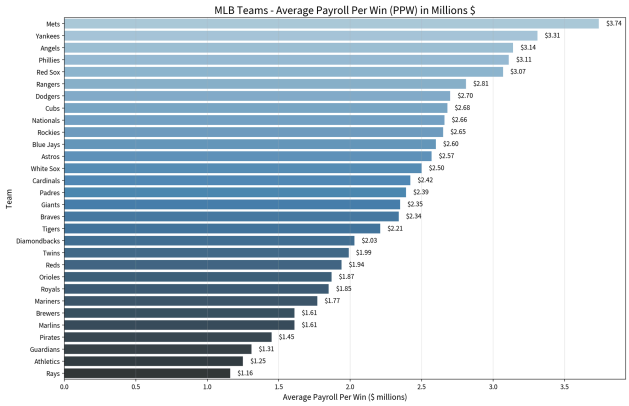
Team	2025	2024	2023	2022	2021	2020	2019	Avg PPW (in millions $)	Owner	Owner NET	Owner gain	WS	Pr 19	pr 20	pr 21	pr 22	pr23	pr 24	pr 25	AVG Payroll (millions)	z score
Angels	2.86	2.79	2.95	2.47	2.36	6.88	1.64	3.14	Arte Moreno	4.9 B	earned		118	179	182	180	215	63	206	$ 163.29	0.45
Astros	2.57	2.77	2.09	1.53	1.82	6.26	0.97	2.57	Jim Crane	2.4 B	earned	Yes	113	213	189	179	201	104	224	$ 174.71	0.53
Athletics	1.04	0.91	1.18	0.85	1.01	2.59	1.20	1.25	John Fisher	3.6 B	inherited		116	101	87	51	59	88	79	$ 83.00	-0.06
Blue Jays	2.48	2.93	2.42	1.90	1.62	3.69	3.13	2.60	Rogers Comm	19 B	inherited		210	118	147	175	215	103	258	$ 175.14	0.67
Braves	2.83	2.61	1.95	1.83	1.47	3.81	1.88	2.34	braves Holdings	2.5 B	earned	Yes	186	164	146	187	205	87	215	$ 170.00	0.49
Brewers	1.24	1.23	1.37	1.60	1.07	3.34	1.40	1.61	Mark attanasio	1.9 B	earned		125	97	103	138	126	104	124	$ 116.71	0.12
Cardinals	1.85	2.20	2.51	1.75	1.87	5.42	1.36	2.42	Bill DeWitt jr	4 B	inherited		124	168	168	163	178	116	144	$ 151.57	0.20
Cubs	2.16	2.75	2.31	1.99	2.11	5.85	1.57	2.68	Tom Ricketts	2.3 B	inherited		132	199	150	147	192	100	207	$ 161.00	0.46
Diamondbac	2.39	1.94	1.32	1.26	1.83	4.48	1.02	2.03	Ken Kendrick	1.2 B	earned		87	112	95	93	124	114	191	$ 116.57	0.39
Dodgers	3.81	3.01	2.36	2.34	2.55	3.96	0.87	2.70	Mark Walter	7.3 B	earned	Yes	94	222	286	262	236	130	404	$ 233.43	1.26
Giants	2.21	2.58	2.47	2.00	1.49	4.86	0.87	2.35	Charles B Johnson	5.5 B	inherited		67	141	162	162	195	144	179	$ 150.00	0.34
Guardians	1.15	1.08	1.29	0.72	0.64	2.83	1.46	1.33	Paul Dolan	4.6 B	inherited		136	99	51	69	98	106	102	$ 94.43	0.03
Mariners	1.74	1.69	1.59	1.28	0.97	3.59	1.54	1.77	John Stanton	2.4 B	earned		105	97	87	118	140	172	167	$ 126.57	0.30
Marlins	0.89	1.71	1.31	1.22	0.94	2.27	2.93	1.61	Bruce Sherman	500 M	earned		167	75	63	84	110	130	70	$ 99.86	-0.10
Mets	4.11	3.50	4.61	2.77	2.65	6.54	2.02	3.74	Steve Cowen	23 B	earned		174	170	204	283	346	173	341	$ 241.57	1.00
Nationals	1.79	1.83	1.54	2.44	2.49	7.00	1.53	2.66	Lerner Family	6.4 B	inherited	Yes	161	182	162	134	109	183	118	$ 149.86	0.10
Orioles	2.22	1.13	0.65	0.76	1.21	2.60	4.48	1.87	David Rubenstein	4.7 B	earned		242	65	63	63	66	184	160	$ 120.43	0.27
Padres	2.35	1.77	3.11	2.25	2.20	4.03	1.01	2.39	Seidler Family	3 B	inherited		78	157	174	214	255	147	214	$ 177.00	0.49
Phillies	3.01	2.58	2.52	2.48	2.41	6.68	2.11	3.11	John Middleton	4.3 B	inherited		171	187	198	243	247	206	292	$ 220.57	0.80
Pirates	1.23	1.14	0.92	0.95	0.79	3.32	1.81	1.45	Bob Nutting	1.1 B	inherited		125	63	48	59	70	248	87	$ 100.00	-0.03
Rangers	2.78	2.91	2.08	2.15	1.63	6.86	1.23	2.81	Ray Davis	3.8 B	earned	Yes	96	151	98	146	214	232	225	$ 166.00	0.53
Rays	1.03	1.10	0.78	0.99	0.70	1.45	2.08	1.16	Patrick Zalupski	1.6 B	earned		208	74	71	85	77	228	79	$ 117.43	-0.06
Red Sox	2.32	2.27	2.55	2.47	1.97	7.38	2.52	3.07	John W Henry	5.7 B	earned		212	177	193	193	199	244	209	$ 203.86	0.47
Reds	1.43	1.27	1.06	1.73	1.58	4.81	1.69	1.94	Bob Castellini	400 M	inherited		127	149	131	107	87	176	119	$ 128.00	0.10
Rockies	3.00	2.41	2.92	2.00	1.59	5.62	1.04	2.65	Dick Monfort	700 M	inherited		74	146	118	136	172	227	129	$ 143.14	0.14
Royals	1.71	1.28	1.63	1.35	1.23	3.50	2.24	1.85	John Sherman	1.3 B	earned		132	91	91	88	91	217	140	$ 121.43	0.19
Tigers	1.71	1.16	1.55	1.97	1.14	4.65	3.26	2.21	Ilitch Family	3.9 B	inherited		153	107	88	130	121	303	156	$ 151.14	0.25
Twins	1.96	1.59	1.76	1.81	1.67	3.69	1.47	1.99	Jim Pohlad	3.8 B	inherited		148	133	122	141	198	328	137	$ 166.71	0.17
White sox	1.42	3.66	2.90	2.38	1.49	3.78	1.88	2.50	Jerry Reinsdorft	2.3 B	earned		135	136	140	193	177	336	85	$ 171.71	-0.04
Yankees	3.40	2.97	3.34	2.44	2.18	6.76	2.06	3.31	Hal Steinbrenner	1.6 B	inherited		222	250	201	249	274	150	296	$ 234.57	0.82

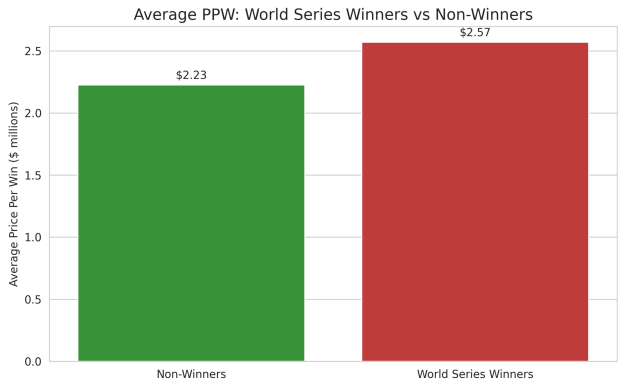
The average MLB team payroll across all 30 franchises is approximately $163.29 million. This figure represents the typical annual player payroll for a Major League Baseball team and serves as the benchmark (mean) for calculating standardized values. The Z-score for this overall average payroll is exactly 0.45, indicating that the league-wide average sits moderately above the long-term or comparative mean used in the analysis (a positive Z-score shows it is higher than the reference point by about 0.45 standard deviations).

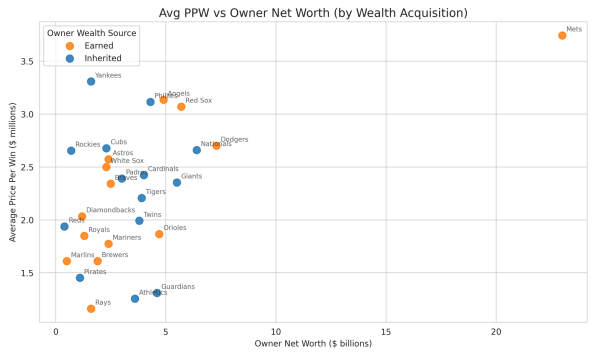
Individual team payrolls differ considerably from this average. For context, high-spending teams such as the Dodgers ($233.43 million, Z-score 1.26), Mets ($241.57 million, Z-score 1.00), Phillies ($220.57 million, Z-score 0.80), and Yankees ($234.57 million, Z-score 0.82) demonstrate substantially above-average investment in player salaries. In contrast, lower-payroll clubs like the Athletics ($83.00 million, Z-score -0.53), Rays ($117.43 million, Z-score -0.06), and Marlins ($98.86 million, Z-score -0.30) fall well below the league average. This distribution illustrates the financial inequalities in MLB, where teams like the Dodgers spend vastly more than historically frugal teams like the Athletics. This growing discrepancy may also lead to a competitive imbalance if the frugal teams spend inefficiently, whereas the Dodgers have more leeway to spend efficiently due to a larger proverbial net.



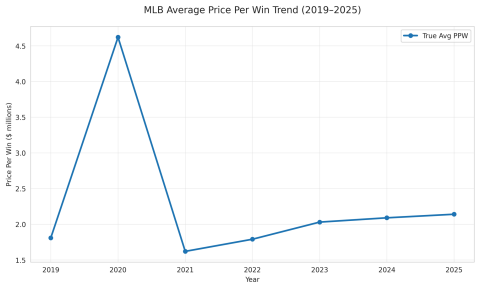
Using methods learned in STOR 320, each team's average PPW was calculated, teams were ranked by average PPW, and owners were categorized as "earned" or "inherited." Three visualizations were generated exclusively from the dataset:

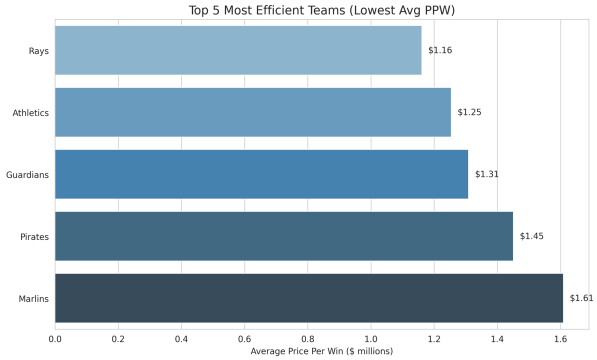


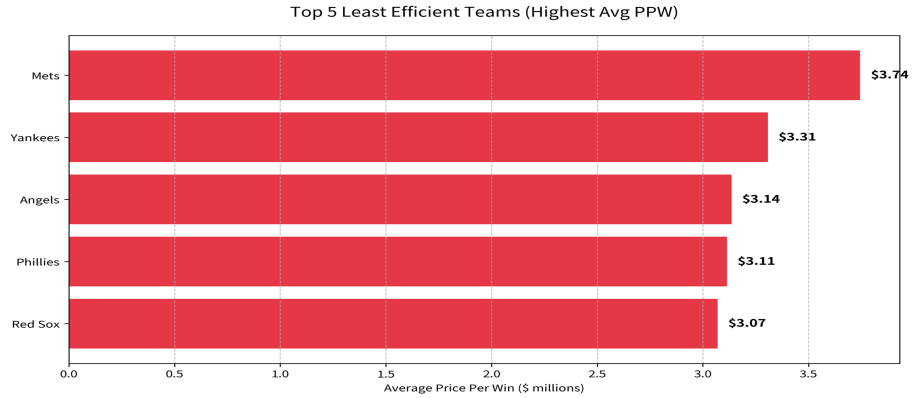




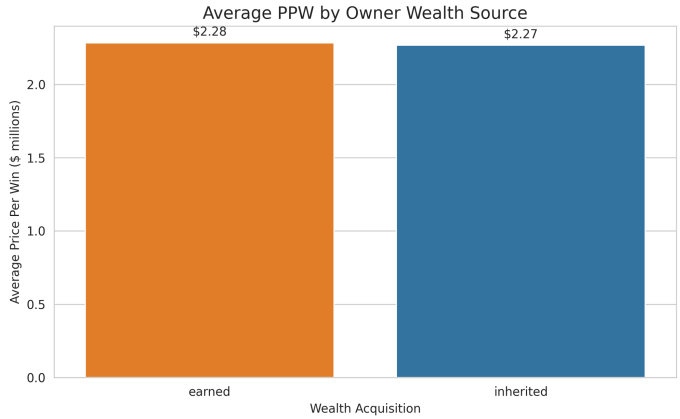
The Tampa Bay Rays recorded the lowest average payroll per win at $1.16 million. The findings illustrate that the low-PPW model produces strong competitiveness, supporting the thesis that spending efficiently is more valuable than spending a large amount. World Series winners from 2019 to 2022 were all below the true average PPW for their respective years, while the past three winners spent above average. This trend suggests the league has shifted toward a “pay-to-win” style at the top in recent seasons, yet the consistently low-PPW teams remain competitive year after year.







Owner wealth patterns show mixed results. Owners who earned their wealth have an average PPW of $2.32 million, while owners who inherited their wealth average $2.29 million, having nearly identical efficiency. The data partially support the hypothesis that earned-wealth owners are more likely to spend more (slightly higher PPW), but it does not show that they spend more efficiently; inherited-wealth owners achieve marginally better efficiency. The highest owner net worth does not guarantee the lowest PPW or most wins.



The findings show that low-PPW spending drives competitiveness even if it does not guarantee championships. Too much reliance on single metrics, such as PPW, without contextual normalization can widen competitive imbalances or lead to flawed owner decisions. This analysis also shows that the Rays pay the least per win and that efficient spending still somewhat remains more valuable than high spending for lasting success. Owner net worth and acquisition method influence spending patterns, but efficiency, rather than raw wealth, better predicts sustained competitiveness. The study highlights the practical power of database normalization, relationship modeling, and visualization within sports analytics.

References

Boyd, D., & Crawford, K. (2012). Critical questions for big data. Information, Communication & Society, 15(5), 662–679.

Elmasri, R., & Navathe, S. B. (2017). Fundamentals of database systems (7th ed.). Pearson.

Floridi, L. (2013). The ethics of information. Oxford University Press.

Han, J., Kamber, M., & Pei, J. (2012). Data mining: Concepts and techniques (3rd ed.). Morgan Kaufmann.

Provost, F., & Fawcett, T. (2013). Data science for business. O'Reilly Media.

Fangraphs.com

BaseballReference.com